Abstract

In the digital age, safeguarding sensitive information has become a critical concern across industries and personal domains. Encryption and decryption are core processes in cryptography that ensure data confidentiality, integrity, and authenticity. Encryption converts plaintext into ciphertext using algorithms and keys, making it unreadable to unauthorized users. Decryption reverses this process, restoring the original data for authorized access.

Modern cryptographic systems employ a variety of algorithms, including:

- Symmetric encryption (e.g., AES – Advanced Encryption Standard), where the same key is used for both encryption and decryption.
- Asymmetric encryption (e.g., RSA – Rivest-Shamir-Adleman), which uses a public key for encryption and a private key for decryption.
- Elliptic Curve Cryptography (ECC), offering strong security with smaller key sizes.

These techniques are widely used in securing:

- Online communications (emails, messaging apps)
- Financial transactions
- Cloud storage
- Digital identity verification

The effectiveness of encryption depends on key management, algorithm strength, and implementation practices. As cyber threats evolve, encryption standards continue to advance, incorporating quantum-resistant algorithms and enhanced performance metrics.

These abstract underscores the pivotal role of encryption and decryption in modern cybersecurity, highlighting their applications, mechanisms, and ongoing relevance in protecting digital assets.